BAPHY105: Engineering Physics

Syllabus Modules Overview

Credits:	3-0-2-4 (3 Lecture, 0 Tutorial, 2 Practical, 4 Credits)
Total Lecture Hours:	45 Hours
Core Focus:	Quantum Mechanics & Elements of Quantum Computing

Syllabus Modules

Module 1 (8 Hours)	**Wave-Particle Duality:** Radiation dual nature, double-slit and Stern-Gerlach experiments, blackbody radiation, Planck and de Broglie hypotheses, and wavefunction interpretation.
Module 2 (10 Hours)	**Mathematical Foundations of Quantum Mechanics:** Linear vector spaces, Hilbert space, Dirac notation, operators (Hermitian, Unitary), eigenvalues, eigenvectors, and Pauli matrices.
Module 3 (8 Hours)	**Postulates of Quantum Mechanics:** Physical observables, expectation values, quantum measurement, sequential Stern-Gerlach setups, and Schrödinger equations.
Module 4 (7 Hours)	**Applications of Quantum Mechanics:** Particles in 1D and 3D potential wells, energy eigenvalues/eigenfunctions, degeneracy, and quantum tunneling.
Module 5 (10 Hours)	**Elements of Quantum Computing:** Introduces qubits, the Bloch Sphere, multi-qubit systems, entanglement, and quantum logic gates like Pauli, Hadamard, and CNOT.
Module 6 (2 Hours)	**Contemporary Topics:** Highlights recent advancements in quantum physics and computing.

Reference Textbooks

1 A. C. Philips, *Introduction to Quantum Mechanics*, Wiley India (2014).
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2 D. J. Griffiths, *Introduction to Quantum Mechanics*.
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